



# PROJECT DOCUMENTATION REPORT

## 1. Project Overview & Objectives

The Smart Agriculture Monitor & Automated Irrigation System is an Internet of Things (IoT) edge solution designed to address water scarcity and unpredictable weather in modern farming. By collecting real-time ecological data, this embedded system enables automated, precision-based watering cycles to save resource costs and optimize soil quality.

### Core Objectives:

- **Automate Moisture Regulations:**  
Execute autonomous water distribution based on real-time soil requirements without human intervention.
- **Environmental Telemetry:** Keep continuous tracks of ambient microclimate shifts (atmospheric temperature and humidity levels).
- **Cloud Integration:**  
Stream localized physical field parameters safely onto an online database infrastructure for analytics.
- **Remote Override Controls:** Provide manual electronic switches to enable remote overrides from anywhere across the globe.

## 2. Comprehensive System Architecture

### Hardware Layer

- **Processing Unit (NodeMCU ESP8266):**  
Operates on a 32-bit Tensilica L106 controller core, handling wireless TCP/IP stacks and executing local algorithmic loops.
- **Atmospheric Node (DHT11):**  
Uses a resistive element for humidity evaluation and an onboard NTC thermistor element to sense ambient heat gradients.
- **Sub-Surface Node (Capacitive Moisture Sensor v1.2):**  
Relies on capacitive changes to judge water volume presence, avoiding oxidation degradation common in cheaper resistive probes.
- **Actuation Module (Single-Channel Relay + DC Pump):** Isolates delicate 3.3V logic circuits from raw 5V motor inductive loads.

## Software & Cloud Layer

- **Embedded Software Layer:**  
Written in C++ using the Arduino framework, running non-blocking interval interrupt arrays to sample peripheral inputs.Cloud Middleware
- **Platform (Blynk IoT):**  
Operates over low-overhead MQTT/Websocket connections to route variables from hardware pins to dynamic frontend endpoints.

## 3. Circuit Connections Mapping Matrix

| Source Component     | Physical Pin | Destination Node | Target Pin | Purpose                             |
|----------------------|--------------|------------------|------------|-------------------------------------|
| DHT11 Sensor         | VCC          | NodeMCU ESP8266  | 3V3 (3.3V) | Main logic power                    |
| DHT11 Sensor         | GND          | NodeMCU ESP8266  | GND        | Common reference ground             |
| DHT11 Sensor         | DATA         | NodeMCU ESP8266  | D4 (GPIO2) | Single-bus digital signal data line |
| Soil Moisture Sensor | VCC          | NodeMCU ESP8266  | 3V3 (3.3V) | Excitation power supply             |
| Soil Moisture Sensor | GND          | NodeMCU ESP8266  | GND        | Common reference ground             |
| Soil Moisture Sensor | A0           | NodeMCU ESP8266  | A0 (ADC0)  | Continuous raw analog reading loop  |
| 5V Relay Switch      | VCC          | NodeMCU ESP8266  | Vin (5V)   | Direct 5V power from USB bus        |
| 5V Relay Switch      | GND          | NodeMCU ESP8266  | GND        | Common reference ground             |

|                 |    |                 |            |                                      |
|-----------------|----|-----------------|------------|--------------------------------------|
| 5V Relay Switch | IN | NodeMCU ESP8266 | D1 (GPIO5) | Relay open/close logic control pulse |
|-----------------|----|-----------------|------------|--------------------------------------|

## 4. Operational Algorithm Flow

[START Setup Sequence]

- Initialize Serial Baud Rate (115200)
- Define Relay Output Pins & Set Logic HIGH (Pump OFF)
- Establish Handshake Connection with Wi-Fi & Blynk Cloud

Network

[ENTER Continuous Execution Loop]

- Trigger Timer Interrupt (Every 2000 milliseconds)
  - Read Ambient Humidity (%) & Temperature (°C) from DHT11
  - Read Analog Value from Soil Probe (A0)
  - Map Raw ADC Value (400 - 850) —> Volumetric Range

(100% - 0%)

- Push Variable Packets to Blynk Virtual Pins (V1, V2, V3)
- [Evaluate Irrigation Automation Rule Thresholds]

[Pump ON]

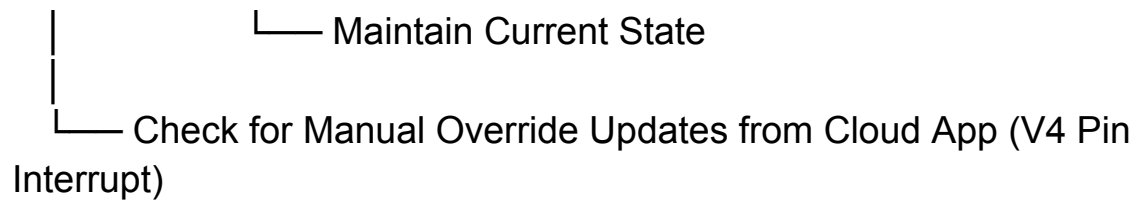
- IF (Soil Moisture < 30%)
  - Drive D1 LOW —> [Engage Relay Module] —>

[Pump OFF]

- ELSE IF (Soil Moisture > 70%)
  - Drive D1 HIGH —> [Disengage Relay Module] —>

[Pump OFF]

- ELSE



## Future Scope

- Integration of AI-based crop prediction models.
- Weather forecasting integration.
- Mobile application development.
- Solar-powered operation.
- Advanced analytics and cloud-based data storage.
- Support for multiple field zones and additional sensors.